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MTH301 Final Term Papers By Waqar (File 4)

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

Answer (Please select your correct option)

VuAnswers.com

- $\int_0^{\frac{\pi}{2}} \cos^n x\,dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\cdots\cdots\frac{5}{6}\frac{3}{4}\frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\,dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\cdots\cdots\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\,dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\cdots\cdots\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\,dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\cdots\cdots\frac{6}{7}\frac{4}{5}\frac{2}{3}$ correct

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4\cos 3x$?

Answer (Please select your correct option)

VuAnswers.com

- 1
- 3
- 4 correct
- 12

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

Answer (Please select your correct option)

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- $\frac{\pi}{2}$
- π correct
- $\frac{3\pi}{2}$
- 4π

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

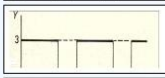
Match the following periodic function with its graph.

$$f(x)=\begin{cases}4 & 0<x<5\\ 0 & 5<x<8\end{cases}$$

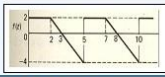
Answer (Please select your correct option)

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

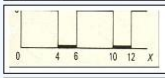
Match the following periodic function with its graph.

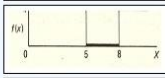
$$f(x)=\begin{cases}4 & 0<x<5\\ 0 & 5<x<8\end{cases}$$

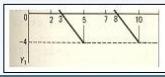
Answer (Please select your correct option)

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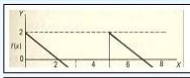
Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

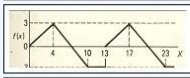
Match the following periodic function with its graph.

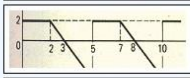
$$f(x)=\begin{cases}2-x & 0<x<3\\ -1 & 3<x<5\end{cases}$$

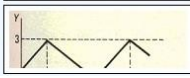
Answer (Please select your correct option)

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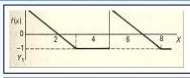
Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

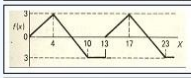
Match the following periodic function with its graph.

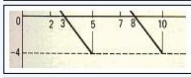
$$f(x)=\begin{cases}2-x & 0<x<3\\ -1 & 3<x<5\end{cases}$$

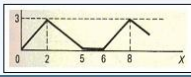
Answer (Please select your correct option)

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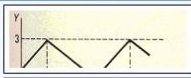
Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

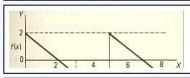
Match the following periodic function with its graph.

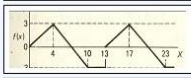
$$f(x)=\begin{cases}\frac{3}{4}x & 0<x<4\\ 7-x & 4<x<10\\ -3 & 10<x<13\end{cases}$$

Answer (Please select your correct option)

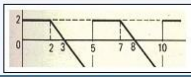
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correct



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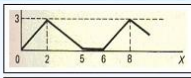
Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

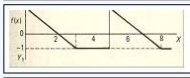
Match the following periodic function with its graph.

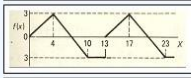
$$f(x)=\begin{cases}\frac{3}{4}x & 0<x<4\\ 7-x & 4<x<10\\ -3 & 10<x<13\end{cases}$$

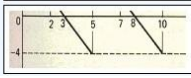
Answer (Please select your correct option)

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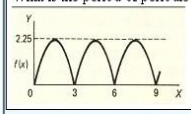
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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

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- 0
- 2.25
- 3 correct
- 6

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^3 e^x$ is ----------

Answer (Please select your correct option)

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- Even function
- Odd function
- Neither even nor odd correct

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about ----------

Answer (Please select your correct option)

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- x-axis
- y-axis correct
- origin

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

Answer (Please select your correct option)

VuAnswers.com

- $3te^{2t} - 2$
- $3e^{2t} - 2t$
- $3e^{2t} - 2$ correct
- None of these.

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

Answer (Please select your correct option)

VuAnswers.com

- $L\{e^{5t}\} = \frac{1}{s-5}$ correct
- $L\{e^{5t}\} = \frac{s}{s^2+25}$
- $L\{e^{5t}\} = \frac{5}{s^2+25}$
- $L\{e^{5t}\} = \frac{5!}{s^6}$

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

Answer (Please select your correct option)

VuAnswers.com

- $L(\sin 3t) = \frac{3}{s^2+9}$ correct
- $L(\sin 3t) = \frac{s}{s^2+9}$
- $L(\sin 3t) = \frac{1}{s-3}$
- $L(\sin 3t) = \frac{3!}{s^4}$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If L denotes laplace transform then
$L(te^{5t}) =$

Answer (Please select your correct option)

VuAnswers.com

- $L(te^{5t}) = \frac{1}{s^2 - 5}$
- $L(te^{5t}) = \frac{1}{s^2 + 5}$
- $L(te^{5t}) = \frac{1}{(s+5)^2}$
- $L(te^{5t}) = \frac{1}{(s-5)^2}$ correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{s}{s^2 + 25}$

Answer (Please select your correct option)

VuAnswers.com

- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \sin 5t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 5t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 25t$

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

Divergence of a vector function is always a ----------

Answer (Please select your correct option)

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- Scalar correct
- Vector

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

If p is the period of a function then that function is said to be periodic if $f(x+p) = f(x)$, _____

Answer (Please select your correct option)

VuAnswers.com

- For all values of x in the domain of f
  correct
- For positive values of x in the domain of f
- For negative values of x in the domain of f

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is associated to each point on a plane?

Answer (Please select your correct option)

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- A real number
  correct
- A natural number
- An ordered pair

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the interval (-1, 5) on co-ordinate line. What does this interval denote?

Answer (Please select your correct option)

VuAnswers.com

- The set of all real numbers between -1 and 5
  correct
- The set of all integers between -1 and 5
- The set of all natural numbers between -1 and 5
- The set of all rational numbers between -1 and 5

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Equation of the circular disk with radius a and origin at $(0,0)$ is given by . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- $x^2 + y^2 = a^2$ correct
- $x^2 + y^2 \geq a^2$
- $x^2 + y^2 \leq a^2$

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

In three dimensional space, the equation $y = x^2$ always represents . . . . . .

Answer (Please select your correct option)

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- Parabola correct
- Straight line
- Half cylinder
- Cone

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Domain of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- Entire 3D-Space correct
- Entire 3D-Space except origin
- First octant of 3D-Space

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = x^3e^{xy}$. Which of the following options is correct?

Answer (Please select your correct option)

VuAnswers.com

- $\frac{\partial f}{\partial y} = 3x^3e^{xy}$
- $\frac{\partial f}{\partial y} = x^3e^{xy}$
- $\frac{\partial f}{\partial y} = x^4e^{xy}$
- $\frac{\partial f}{\partial y} = x^3ye^{xy}$ correct

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a . . . . . function.

Answer (Please select your correct option)

VuAnswers.com

- Scalar
- Continuous
- Vector correct
- Constant

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of a rectangular box. The area of bottom will be . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- $A = yz$
- $A = xz$
- $A = xy$ correct
- $A = xyz$

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of an open rectangular box. The surface area of the box will be . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- $A = xy + 2yz + 2xz$
- $A = yz + 4$
- $A = xz + yz + zx$
- $A = xyz$

correct

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 0 \le x \le 2 \text{ and } 0 \le y \le 3\}$, then $\iint_R (1 - ye^{xy}) dA =$ . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- $\int_0^2 \int_0^3 (1 - ye^{xy}) dy dx$

correct

- $\int_0^2 \int_0^3 (1 - ye^{xy}) dx dy$
- $\int_2^3 \int_0^0 (1 - ye^{xy}) dx dy$
- $\int_0^2 \int_2^3 (4xe^{2y}) dy dx$

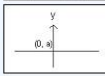
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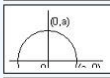
Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.
$r \sin\theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

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- 
- 

- 

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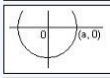
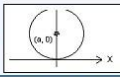
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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.
$r \sin\theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

VuAnswers.com

- 
- 
- 
- 

correct

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $(-8, 45^0)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

VuAnswers.com

- $(8, 225^0)$
- $(-8, 225^0)$
- $(8, 45^0)$
- $(8, 315^0)$

correct

Made by: Waqar Siddhu

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

VuAnswers.com

- $\left(7, \frac{3\pi}{4}\right)$
- $\left(-7, \frac{3\pi}{4}\right)$
- $\left(-7, \frac{-\pi}{4}\right)$
- $\left(7, \frac{-3\pi}{4}\right)$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If $p(r,\theta)$ is a point in polar coordinate system, then r is the distance of p from . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- Pole — correct
- Imaginary axis
- None of these
- Polar axis

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . . where $a \le \theta \le b$ and $c \le r \le d$.

Answer (Please select your correct option)

VuAnswers.com

- $\int_a^b \int_c^d f(r,\theta)rdrd\theta$ — correct
- $\int_a^b \int_c^d f(r,\theta)drd\theta$
- $\int_a^b \int_c^d f(r,\theta)rd\theta dr$
- $\int_a^b \int_c^d f(r,\theta)d\theta dr$

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of the equation $r = 3\hat{i} - 2\hat{j} - \hat{k}$ is the point . . . . . .

Answer (Please select your correct option)

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- $(1,-2,3)$
- $(3,-2,-1)$ — correct
- $(3,2,1)$
- $(-3,-2,-1)$

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The natural domain of a vector valued function is the . . . . . of the natural domains of its components

Answer (Please select your correct option)

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- Product
- Summation
- Union

correct

- Intersection

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\vec{R}(t)$ is the anti-derivative of a given vector valued function $\vec{r}(t)$ i.e., $\vec{R}'(t) = \vec{r}(t)$ then $\int_a^b \vec{r}(t)dt =$ . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- $\vec{R}(a) + \vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a) + c$ where c is a non-zero constant.

correct

- $\vec{R}(a).\vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a)$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t) \times \vec{r}_2(t)] =$ . . . . . where "×" and "." represent the cross and dot product respectively.

Answer (Please select your correct option)

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- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \vec{r}_2 \times \frac{d\vec{r}_1}{dt}$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector valued function $\vec{F}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is continuous if . . . . . .

Answer (Please select your correct option)

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- Atleast one of its components is continuous.
- All of its components are necessarily differentiable. correct
- All of its components are continuous.
- Limit exists for all of its components.

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Given a vector valued function $\vec{F}(t) = \frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ and its anti-derivative $\vec{R}(t) = \ln(t-3)\hat{i} + e^t\hat{j}$, then $\int \vec{F}(t)dt =$ . . . . . .

Answer (Please select your correct option)

VuAnswers.com

- $\ln(t-3)\hat{i} + e^t\hat{j} + c$
- $(t-3)\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $(t-3)^{-1}\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $\frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ correct

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t).\vec{r}_2(t)] =$ . . . . . where "×" and "." represent the cross and dot product respectively.

Answer (Please select your correct option)

VuAnswers.com

- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \times \vec{r}_2$ correct

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A single curve can be represented by . . . . . vector valued function(s).

Answer (Please select your correct option)

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- Two
- Infinitely many
- Single

correct

- Three

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

A function is said to be smooth if it's derivative is . . . . . on any value of its domain.

Answer (Please select your correct option)

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- continuous and non zero
- piecewise continuous
- defined and non zero
- differentiable

correct

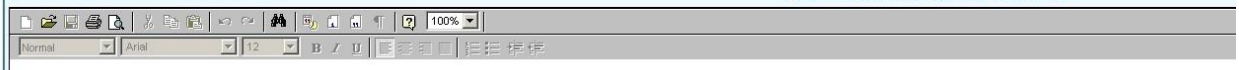
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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$

Answer (Please click here to Add Answer)

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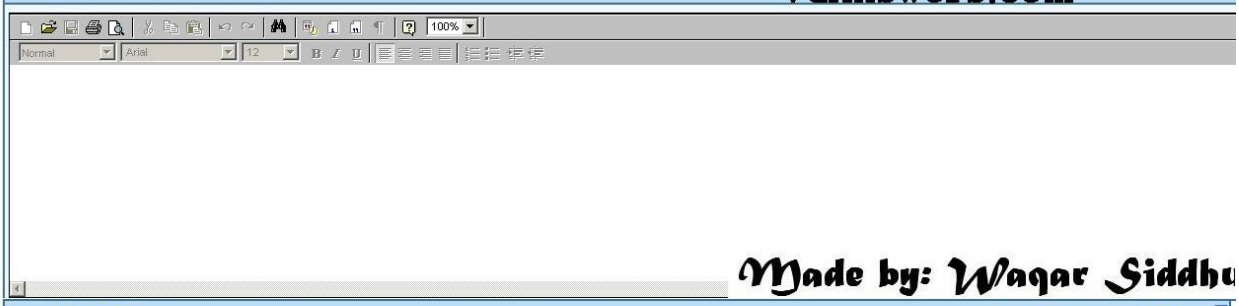
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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Find Laplace transform of the function $F(t)$ if $F(t) = \cos 3t$.

Answer (Please click here to Add Answer)

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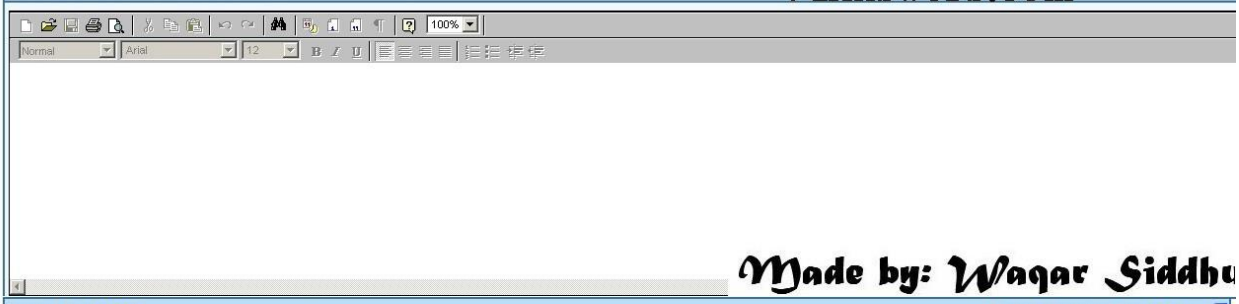
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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

State the condition when $\iint_R f(x,y)dA = \int_c^d \int_a^b f(x,y)dxdy = \int_a^b \int_c^d f(x,y)dydx$ where R is the region of integration.

Answer (Please click here to Add Answer)

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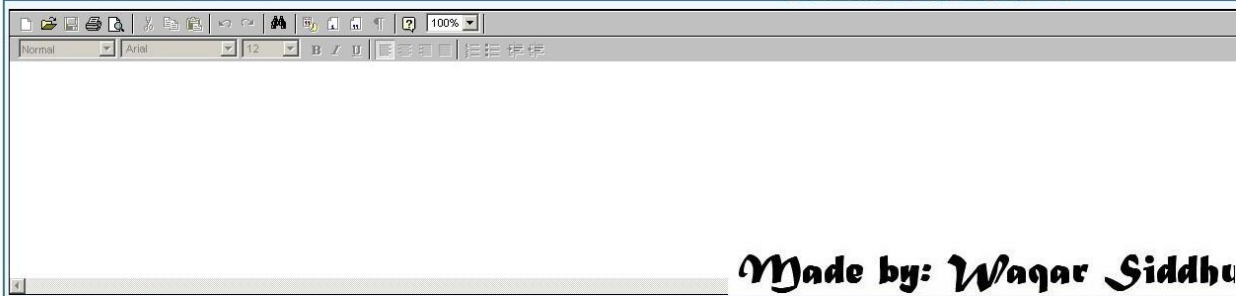
Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Find derivative of the following vector-valued function.

$\vec{r}(t) = e^{t^2}\hat{i} + t^2\hat{j} + \sec 2t\,\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^4 x\right) dx$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find Laplace transform of the function $F(t)$ if
$F(t) = e^{2t} \sin 3t$

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the critical point for the given function $f(x, y) = 6x^2 + xy - 2y^2$ along the line $y = 3x + 1$ at which the absolute extrema of the function can occur.

Answer (Please click here to Add Answer)

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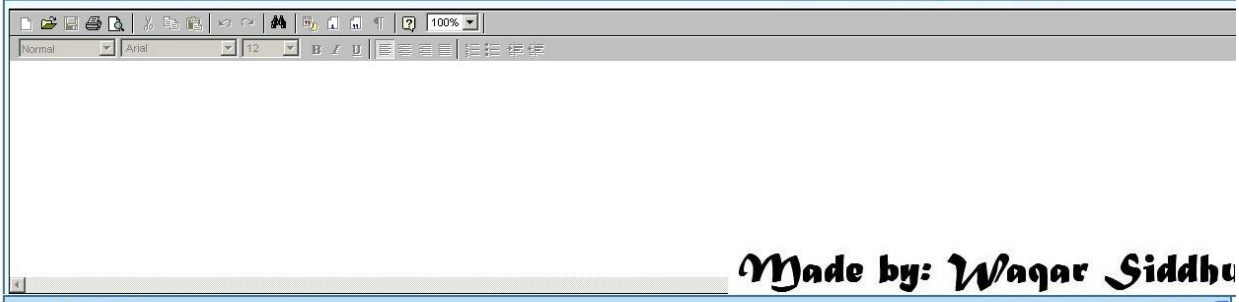
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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ when $0 \le t \le 2\pi$?

Answer (Please click here to Add Answer)

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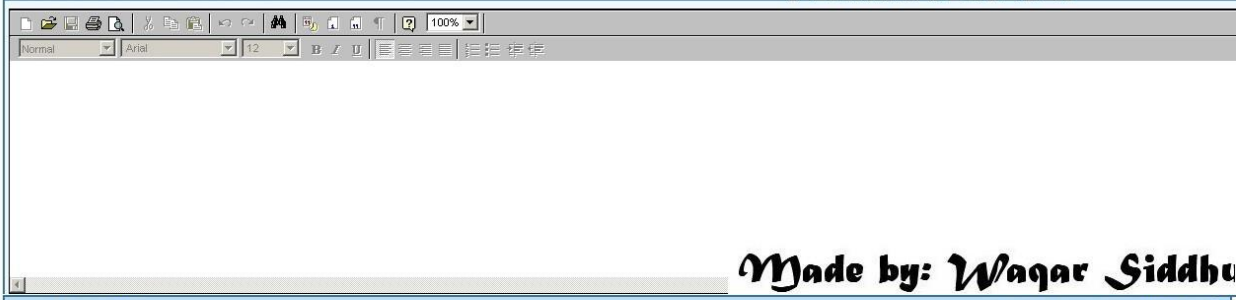
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine the Fourier co-efficient b_n of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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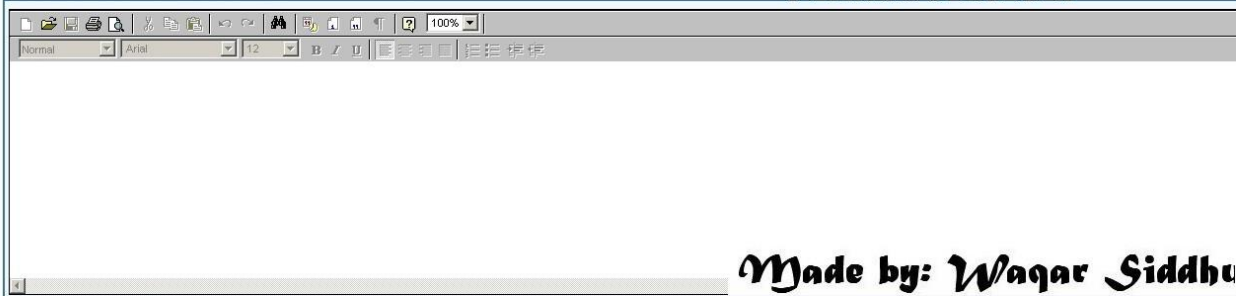
Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = x^2 z\,\hat{i} + y^2 x\,\hat{j} + (y + 2z)\,\hat{k}$

Answer (Please click here to Add Answer)

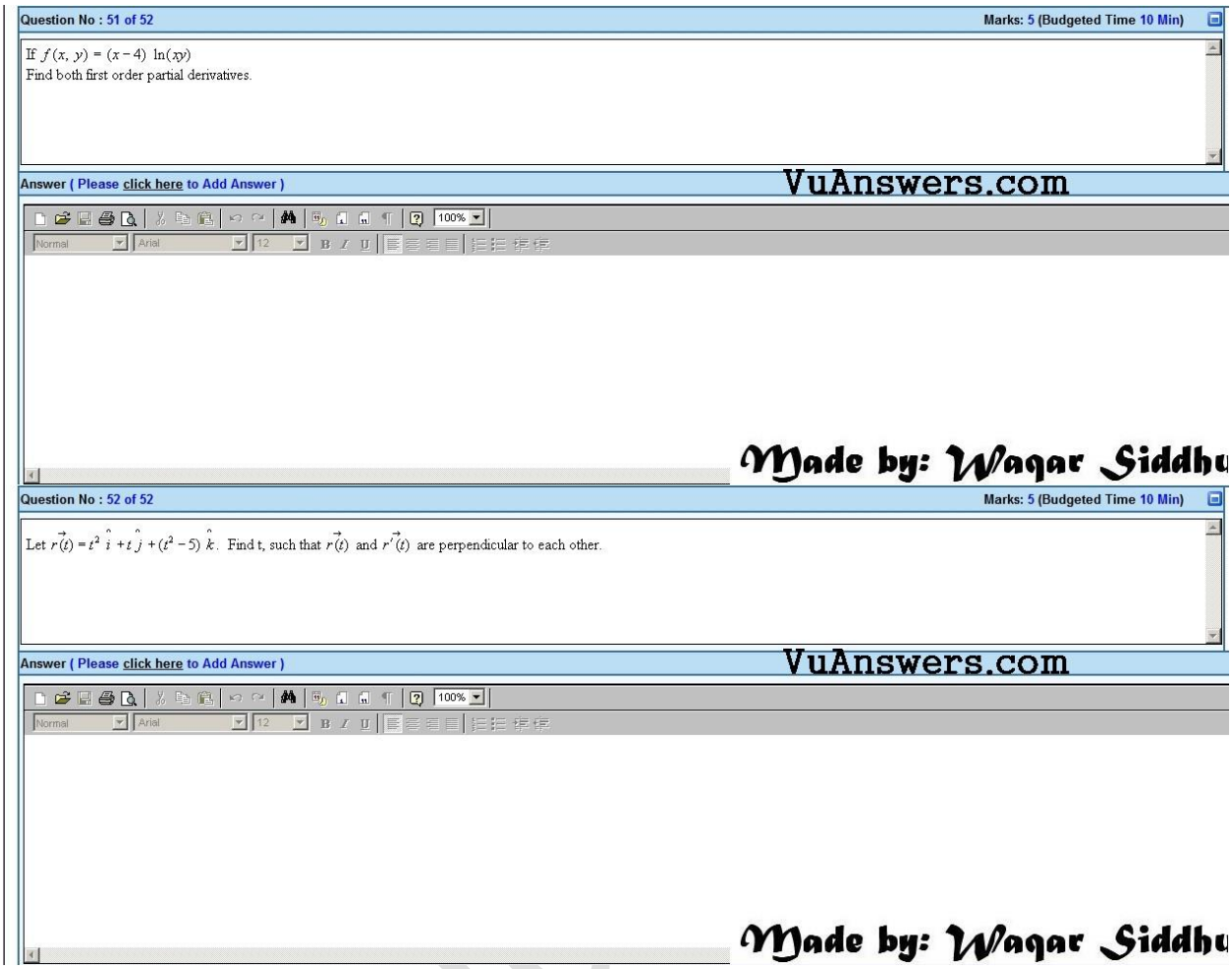
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